

Kubernetes

Complete Installation & Setup

Beginner guide - from download to first use

Objective

By the end of this manual, you should be able to install the tool, validate the installation, perform the first functional use and recognize common errors. Do not skip validation steps: they confirm that each stage actually worked.

Before you start

For command-line access install kubectl. A local cluster additionally requires a solution such as Docker Desktop Kubernetes, minikube or kind.

Official download and documentation

<https://kubernetes.io/docs/setup/>

Attention

Use only test environments, accounts and devices when a tool can capture traffic, run scans, modify infrastructure or access data. In corporate environments, follow your organization policies.

Open the correct terminal

On Windows, use CMD or PowerShell. On macOS and Linux, use Terminal. Run commands exactly as shown and press Enter after each line.

Windows - step by step

Install kubectl using the official Windows instructions and optionally enable/create a local cluster.

```
kubectl version --client
```

macOS - step by step

Install kubectl using the official macOS instructions and optionally create a local cluster.

```
kubectl version --client
```

```
kubectl config current-context
```

Linux - step by step

Install kubectl using the official repository/binary instructions and configure access to a cluster.

```
kubectl version --client
```

```
kubectl config current-context
```

Validate the installation

1. kubectl version --client

2. kubectl config current-context

```
kubectl version --client
```

```
kubectl config current-context
```

Commands and configuration

1. kubectl version --client

```
kubectl version --client
```

Run it in the indicated terminal, read the complete output and resolve any error before the next step.

PATH, environment variables and a new terminal

When an installer adds an executable to PATH or you configure a variable such as JAVA_HOME/ANDROID_HOME, close the old terminal and open a new CMD/PowerShell. Then repeat the version command. If multiple versions are installed, use where <command> on Windows or which <command> on macOS/Linux to identify the executable being used. Do not remove PATH entries without understanding their origin.

Create a safe test folder

For the first tests, use a simple folder with no production files, for example C:\Projects\qa-lab or ~/qa-lab. This makes generated files, virtual environments, logs and configuration easier to identify.

First use / functional test

1. kubectl get nodes

```
kubectl get nodes
```

Expected result

kubectl reports its client version; with a configured cluster, kubectl get nodes returns the cluster nodes.

Troubleshooting

- Read the terminal error before repeating commands. Check PATH/environment variables, runtime version, permissions, system architecture and whether a new terminal was opened after environment changes.
- Command not found: confirm the installation directory is on PATH and open a new terminal.
- Wrong version: check whether another older runtime or executable appears earlier on PATH.
- Permission denied: use a user-writable project directory and follow the vendor permission guidance; do not blindly run everything as administrator/root.
- Corporate proxy/firewall: configure the package manager/tool according to your organization network policy rather than disabling security controls.

Final checklist

- Official download and documentation
- Windows - step by step / macOS - step by step / Linux - step by step
- Validate the installation
- First use / functional test
- Expected result
- Troubleshooting

Installation validated. Keep this manual for reinstallations, onboarding and troubleshooting.